

PROJECT VISION DECK

# AutoRAG

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## Automated RAG Optimization Pipeline

End-to-end agent that builds, evaluates, improves, and deploys RAG systems automatically.

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# What this project is about

AutoRAG is designed to solve one painful problem:

***Building a good RAG system is slow, manual, and guesswork-heavy.***



What is a RAG system?

- "Chat with your documents"
- Internal company knowledge bots
- AI assistants that answer using your data



The Problem Today

- Manual tuning of chunking, embeddings, retrieval, prompts
- No clear scoring system
- No automation
- No reliable way to know "this is good"

It works like this:

1. Search relevant documents

2. Feed them to an AI model

3. Generate an answer grounded in those documents



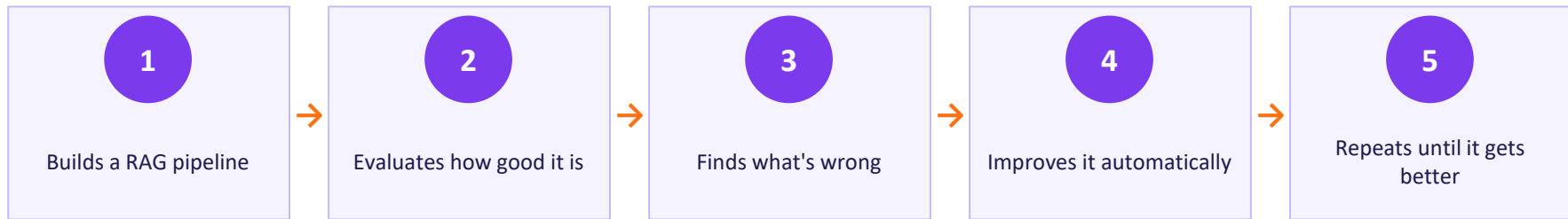
Engineers repeat: Try → test → tweak → repeat (for days)

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# What AutoRAG Does

**AutoRAG turns that manual process into an automated loop**



 **The core idea (very important): AutoRAG is a closed-loop system**



 **No manual tuning required.**

# How It Works Internally

It uses multiple 6 agent cards, each with a job:



## Ingestion

Prepares your documents



## Builder

Creates the RAG pipeline



## Evaluator

Tests it with auto-generated questions



## Diagnoser

Finds why it failed



## Improver

Tries better configurations



## Deployer

Puts the best version into production

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# How It Measures Quality

## The Core Scoring Formula

$$\text{Score} = 0.25 \cdot \text{Retrieval} + 0.35 \cdot \text{Quality} + 0.25 \cdot \text{Faithfulness} - 0.10 \cdot \text{Latency} - 0.05 \cdot \text{Cost}$$



### Retrieval

25%

Did it fetch the right information?



### Quality

35%

Is the answer actually good?  
Clarity, completeness,  
relevance.



### Faithfulness

25%

Is it grounded in documents?  
No hallucinations.



### Latency

-10%

How fast is the response?  
Slower pipelines are  
penalized.



### Cost

-5%

How expensive is it to run?

Score: 0.85+ → Production-ready 0.70–0.84 → Needs improvement < 0.70 → Poor quality

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# Why This Matters

AutoRAG removes the biggest bottleneck in AI systems:

## ✗ BEFORE

- ✗ Manual experimentation
- ✗ Need for labeled data
- ✗ Trial-and-error tuning



## ✓ AFTER

- ✓ Automated optimization
- ✓ Measurable quality
- ✓ Fast iteration (minutes instead of days)



## Who It's For

Engineers building AI products

Data scientists running experiments

Companies deploying internal knowledge bots

Developers who want fast, reliable RAG systems

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## One-Line Summary

**AutoRAG is an automated system that builds and continuously improves AI document-based question-answering systems without manual tuning or labeled data.**

# The Scoring System — Deep Dive

*The scoring in AutoRAG decides: “Is this RAG pipeline good or bad?”*

## Retrieval (25%)

Did the system fetch the right information? Checks if relevant document chunks were retrieved. If retrieval is bad, everything else fails.

## Quality (35%) — Most important

Is the answer actually good? Clarity, completeness, relevance. Measured using an LLM acting as a judge.

## Faithfulness (25%)

Is the answer grounded in the documents? No hallucinations, no made-up facts. Critical for enterprise trust.

## Latency (–10%)

How fast is the response? Slower pipelines are penalized. Encourages real-world usability.

## Cost (–5%)

How expensive is it to run? API usage (LLMs, embeddings), infrastructure cost.

 **Simple way to think about it: Score = How correct + how truthful + how useful – how slow – how expensive**

## Final Score Behavior

Score: 0.0 → 1.0 Higher = better pipeline

0.85 → Production-ready

0.70 → Needs improvement

0.50 → Poor quality

## How it's used in the system

→ Evaluator → calculates it

→ Diagnoser → explains why it's low

→ Improver → tries to increase it

→ Experiment Tracker → compares runs

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# Real Example: How a Pipeline Gets Scored

## Scenario: RAG system on company HR documents

User question: “What is the maternity leave policy?”  
Ground truth: “Employees are entitled to 26 weeks of paid maternity leave.”

| Component                                                                                         | Raw Score | Weight | Contribution |
|---------------------------------------------------------------------------------------------------|-----------|--------|--------------|
|  Retrieval       | 0.70      | ×0.25  | +0.175       |
|  Quality         | 0.65      | ×0.35  | +0.2275      |
|  Faithfulness    | 0.60      | ×0.25  | +0.15        |
|  Latency penalty | 0.20      | ×0.10  | -0.020       |
|  Cost penalty    | 0.10      | ×0.05  | -0.005       |

 **Final Score ≈ 0.53 → Poor quality → System triggers improvement**

### After Improvement

New answer: “Employees are entitled to 26 weeks of paid maternity leave.”

|               |      |
|---------------|------|
| Retrieval:    | 0.85 |
| Quality:      | 0.90 |
| Faithfulness: | 0.95 |

**Final Score ≈ 0.82  Production-ready**

 **Diagnosis: Partial Answer + Mild Hallucination →  Actions: Increase top\_k, stricter prompt, lower temperature**

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## Key Takeaway

**This scoring system doesn't just say “right or wrong.”**



Did we retrieve the right info?



Did the model answer well?



Did it hallucinate?



Is it fast and cheap enough?



**And then uses that to automatically improve the pipeline**